XCITY AI COMPETITION · PARCELS 9 + 13 · SAN JUAN, ARGENTINA

Terrace Oasis: a city that grows its own shade beside the first fields



city-evening · ChatGPT image generation (chatgpt.com, web app, October 2026) from renders/city-evening.png · illustrates the geometry, not evidence

The scheme

Reference example, not an entry. The first city stands on the flat terrace directly east of the pilot area, between the fields and a 30 m bluff, facing the Andes. About 41,340 people live in low courtyard houses on a 300 m grid; every street is an acequia street, shaded by trees watered by gravity from a tank on the bluff; machines keep to their own 600 m grid. Across the site, four oasis cities hang on one corridor, the Line, from the RN 20 gate to the port in the east. The city gives up growth speed, not water.

41,340	**7,891**	**40**	**410**
residents, phase one	jobs in the city	MW of compute	MWp of PV
3,599,292	**100**	**36**	**30**
m³ of water a year	% hourly carbon-free	km² master design	human-machine interfaces

Key nodes: N01 Entrance and arrival sequence · N02 A future housing prototype · N10 Public space in heat and Zonda wind

Declared trade-off: Growth speed: the city does not grow faster than its measured water allows

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

When machines take over much of the work, a desert city should be built around being outside together: shade and water first, and growth only as fast as the water allows.

What it gives up

Growth speed: the city does not grow faster than its measured water allows. Nobody knows the aquifer's recharge. A city sized on hope is the one irreversible mistake a desert city can make.

Who carries the cost

The developer and investors, through slower returns, and households on the waiting list; not the aquifer and not the people already living here.

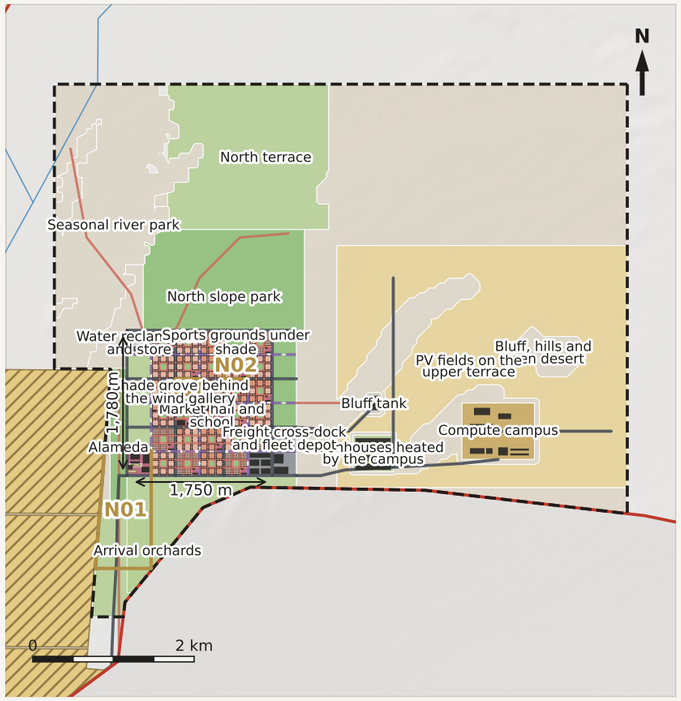
Three key nodes

N01 Entrance and arrival sequence - Arriving along the first fields
N02 A future housing prototype - A superblock of courtyard houses
N10 Public space in heat and Zonda wind - Plaza del Agua: a square for 45 degrees and the Zonda

1 The whole site · structure



2 The master design · drawn by the entrant



3 N01 Entrance and arrival sequence



N02 A future housing prototype



N10 Public space in heat and Zonda wind



Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

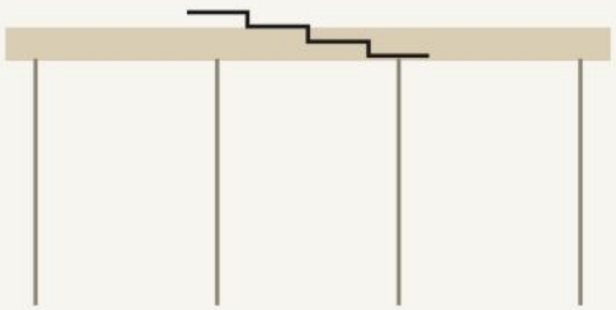
Agent task 1 · drawn by the agent with tools/make_pages.py



Terraza

the water comes down our street, and the street is ours

THE MARK IN USE



On a shade canopy, cut into the fascia



On a delivery robot's status light



Stamped on an olive-oil tin

NAMING · IN WATER ORDER

Terraza

the city

Avenida del Agua

the main street, from the tank on the bluff

Acequia 1, 2, 3 …

the acequia streets, numbered downhill

Acequia 4 · Patio 2

an address: how near you are to the water

Plaza del Agua

the main square

La Llegada

the arrival, under the coach canopy

COLOUR DOES A JOB

land and buildings

water and planting

machine territory

One sans-serif family in two weights. Signs sit on the ground, not on poles. Paving changes texture 20 m before every machine crossing.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Scale 2 · rendered from the geometry



master-aerial · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view master-aerial · illustrates the geometry, not evidence



city-evening · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view city-evening · illustrates the geometry, not evidence

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Every mandatory requirement and agent task

Twelve mandatory requirements (M1-M12), phase one at full occupancy

	Mandatory requirement	This scheme	Evidence	Answered in
M1	Hourly carbon-free energy match	100 %	assumed	§3. Energy and compute · 能源与算力
M2	Peak IT load in phase one	40 MW	declared	§3. Energy and compute · 能源与算力
M3	Annual water demand, by use	3,599,292 m3/yr	assumed	§4. Water · 水
M4	Cooling water intensity (WUE)	0.20 L/kWh	assumed	§4. Water · 水
M5	Response to a 30% water shortfall	stated in the text	-	§4. Water · 水
M6	Human-machine at-grade interfaces	30	computed	§5. Human and machine · 人与机器
M7	Machine traffic rule	combined	declared	§5. Human and machine · 人与机器
M8	Public audit mechanism	stated in the text	-	§8. Governance and data · 治理与数据
M9	Control and succession of the [...]	stated in the text	-	§8. Governance and data · 治理与数据
M10	Triggers for phases two and three	stated in the text	-	§9. Phasing · 分期
M11	Irreversible decisions	stated in the text	-	§9. Phasing · 分期
M12	Jobs and residents reconcile	7,891 jobs · 41,340 persons [...]	declared	§6. Programme and production · 功能与产业

Eight agent tasks

	Agent task	Answered in
A1	Identity of the place	§1. Position · 定位
A2	Energy and compute	§3. Energy and compute · 能源与算力
A3	Water in a dry place	§4. Water · 水
A4	Human and machine	§5. Human and machine · 人与机器
A5	Programme and production	§6. Programme and production · 功能与产业
A6	Urban form across parcels 9 and 13	§7. Urban form · 城市形态
A7	Data and AI governance	§8. Governance and data · 治理与数据
A8	Phasing in three steps	§9. Phasing · 分期

How to read this page

Values come from metrics.json with their evidence tags (figure 07); sections are the headings of proposal.md. The self-check (gate G6) confirms each answer points at a real heading; it does not judge the answer.

What each answer must state (brief/requirements.json)

M1 generation mix, storage duration, and whether the match is physical (self-supplied) or contractual
M2 the grid connection capacity you assumed (not supplied by the organiser)
M3 Phase one at full occupancy. The breakdown must add up to the total (2% tolerance); say which line panel-cleaning water sits in.
M4 cooling type: air, liquid, hybrid or dry - and the PUE that goes with it
M5 30% short means 30% less water than your phase-one annual demand (M3). Give a priority order of what stops first, and what happens in the city at each step.
M6 Count every place where a people or shared route meets a machine route on the same level: road crossings, the ends of shared streets, kerbs or loading points that people and machines share, stops of driverless shuttles on a [...]
M7 where each rule applies and how it is enforced
M8 who publishes power, water, generation and compute data, how often, and who can check it independently
M9 who owns and operates it, and what happens if that operator leaves
M10 observable conditions, not dates
M11 at least three phase-one decisions that cannot be changed later, and why that is acceptable
M12 if residents far exceed jobs, who the others are and what they live on

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Scale 1 · structural



Legend

- Reserve 899 km²
- Agri-production 642 km²
- Energy + compute 58 km²
- Industry + automation 30 km²
- Living + public life 21 km²
- Trade + port 3.30 km²
- Site boundary (1,656 km²)
- Pilot area - fixed, not designed
- Master design area
- Seasonal channels (DEM-derived)
- Key nodes

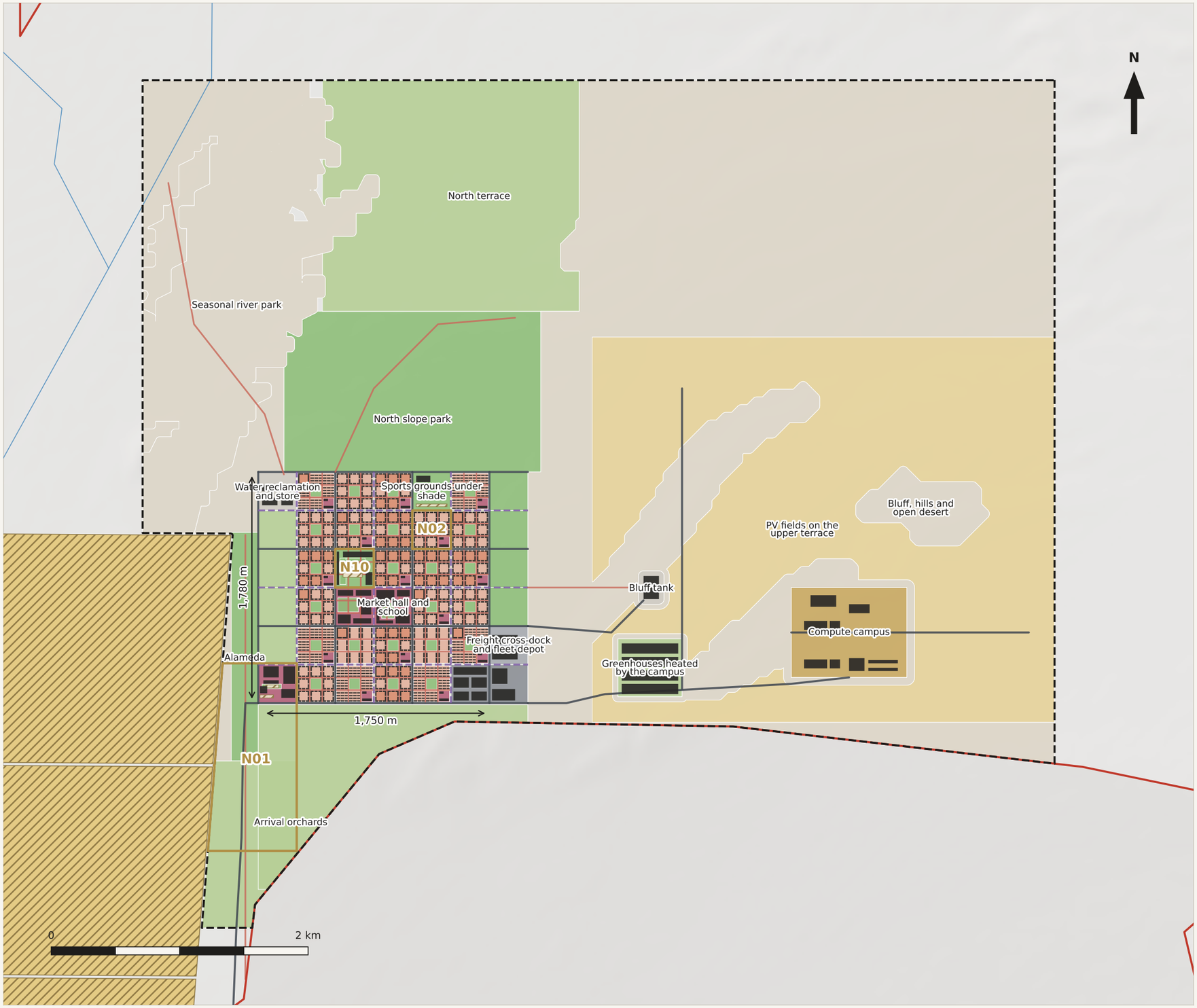
Why the structure sits where it does

Reference example, not an entry. The first city stands on the flat terrace directly east of the pilot area, between the fields and a 30 m bluff, facing the Andes. About 41,340 people live in low courtyard houses on a 300 m grid; every street is an acequia street, shaded by trees watered by gravity from a tank on the bluff; machines keep to their own 600 m grid. Across the site, four oasis cities hang on one corridor, the Line, from the RN 20 gate to the port in the east. The city gives up growth speed, not water.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Scale 2 · figure-ground and land use



Legend

- Agriculture
- Civic
- Compute
- Energy
- Industry
- Infrastructure
- Logistics
- Mixed use
- Open space
- Reserve
- Residential
- Buildings (figure-ground)
- Master design area
- Key node
- Seasonal channel

Master design area

36 km² · drawn by the entrant

Recomputed

Site coverage 2.90% · FAR 0.06 · GFA 2,309,984 m²

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Floors from buildings.geojson; metrics recomputed



Legend

- 1 floor
- 2-3 floors
- 4-6 floors
- 7+ floors
- Master design area

Land use in the master area (ha)

reserve 1,666
energy 782
agriculture 586
open space 361
residential 102
compute 63
mixed use 38
civic 38
industry 18
infrastructure 14
logistics 9.01

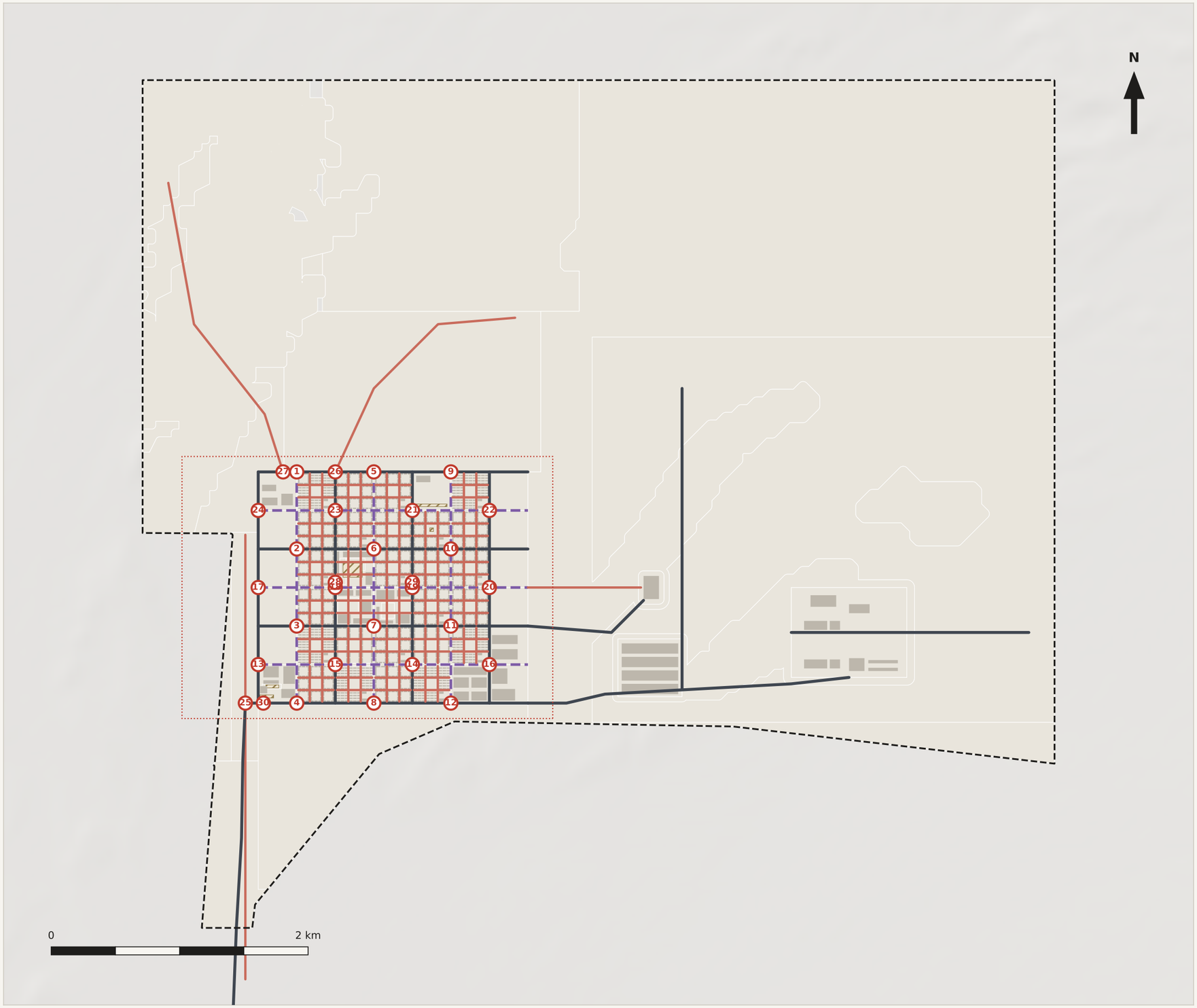
Recomputed from geometry

Coverage 2.90% · FAR 0.06 · open space 9.90%

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Every human-machine interface numbered



Legend

- People 40 km
- Machines + goods 30 km
- Shared 12 km
- Human-machine interface (30)

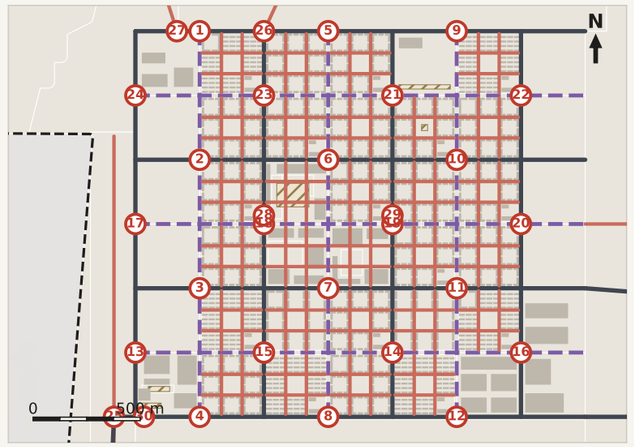
Machine traffic rule

combined

Interfaces

30 at-grade points where people and machines meet

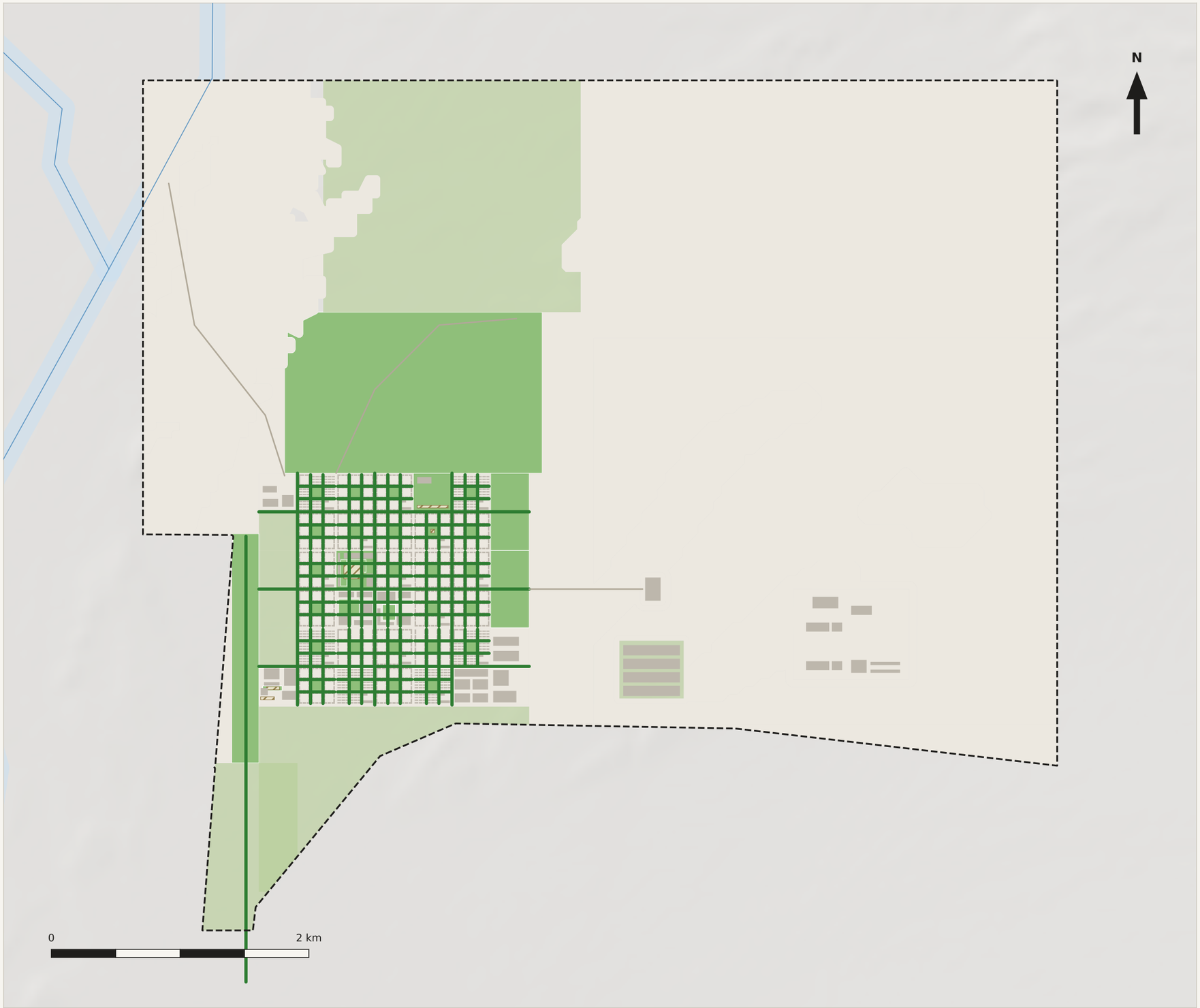
Interfaces, enlarged



Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Channels are DEM-derived flow lines

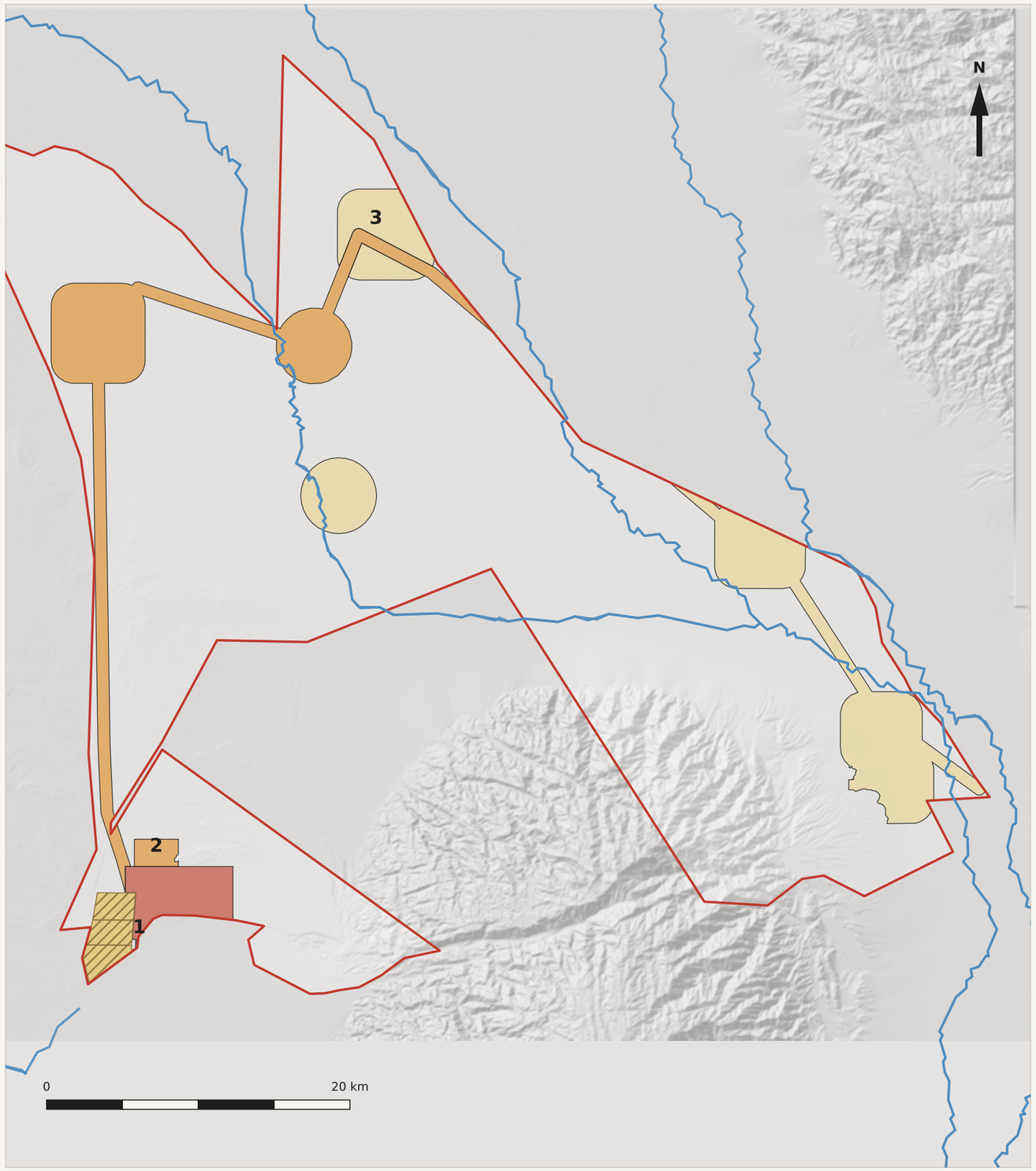


Legend
- Open space 9.90%
- Agriculture
- Shaded walking 90%
- Unshaded walking
- Seasonal channel
- Flood corridor shown at 250 m / 100 m

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Observable triggers, not dates



Legend

- Phase 1 24 km²
- Phase 2 109 km²
- Phase 3 122 km²

Phase 1 - Terrace Oasis: the first city, its fields, PV and campus, and the arrival road

Phase one: no trigger.

Phase 2 - Upper town on the north terrace; the Line north to Oasis 2; second campus

Lower terrace above 85% occupied for 12 consecutive months AND wellfield drawdown below 0.5 m a year for 24 months AND a second abstraction licence granted AND the open-data office has published 24 monthly water and power accounts

Phase 3 - East: Oasis 3 and 4, port of entry, gateway industry, the Line to RP 510

Access corridor to RP 510 financed AND committed export throughput above 40,000 t a year AND phase-two population above 40,000

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Every declared figure with its evidence tag

Metric	Declared	Unit	Evidence	Basis / check
master_area_km2	36	km2	computed	recomputed 36
site_coverage_pct	2.92	%	computed	recomputed 2.92
gfa_m2	2,309,984	m2	computed	recomputed 2,309,984
far	0.06	ratio	computed	recomputed 0.06
open_space_pct	9.91	%	computed	recomputed 9.91
human_machine_interfaces	30	count	computed	recomputed 30
shaded_pedestrian_pct	89	%	computed	recomputed 89
structure_km2	per system	km2	computed	recomputed: per system
hourly_cfe_match_pct	100	%	assumed	A-07
compute_availability_pct	85	%	assumed	A-07
peak_it_load_mw	40	MW	declared	
pv_capacity_mwp	410	MWp	declared	
storage_mwh	1,000	MWh	declared	
pue	1.25	ratio	assumed	A-04
wue_l_per_kwh	0.20	L/kWh	assumed	A-04
waste_heat_reuse_pct	9	%	assumed	A-13
annual_water_demand_m3	3,599,292	m3/yr	assumed	A-01
water_domestic_m3	1,961,583	m3/yr	assumed	A-08
water_compute_m3	59,568	m3/yr	assumed	A-04
water_agriculture_m3	1,218,141	m3/yr	assumed	A-09
water_landscape_m3	300,000	m3/yr	assumed	A-09
water_industry_m3	60,000	m3/yr	assumed	A-09
water_reuse_pct	90	%	assumed	A-08
machine_traffic_rule	combined	enum	declared	
jobs_phase1	7,891	jobs	declared	
residents_phase1	41,340	persons	declared	
dwellings_phase1	17,224	units	declared	
long_term_residents	160,000	persons	declared	

Evidence tags

- computed - recomputed from your geometry
- sourced - cited in sources.json
- assumed - declared in assumptions.json
- indicative - organiser's indicative material

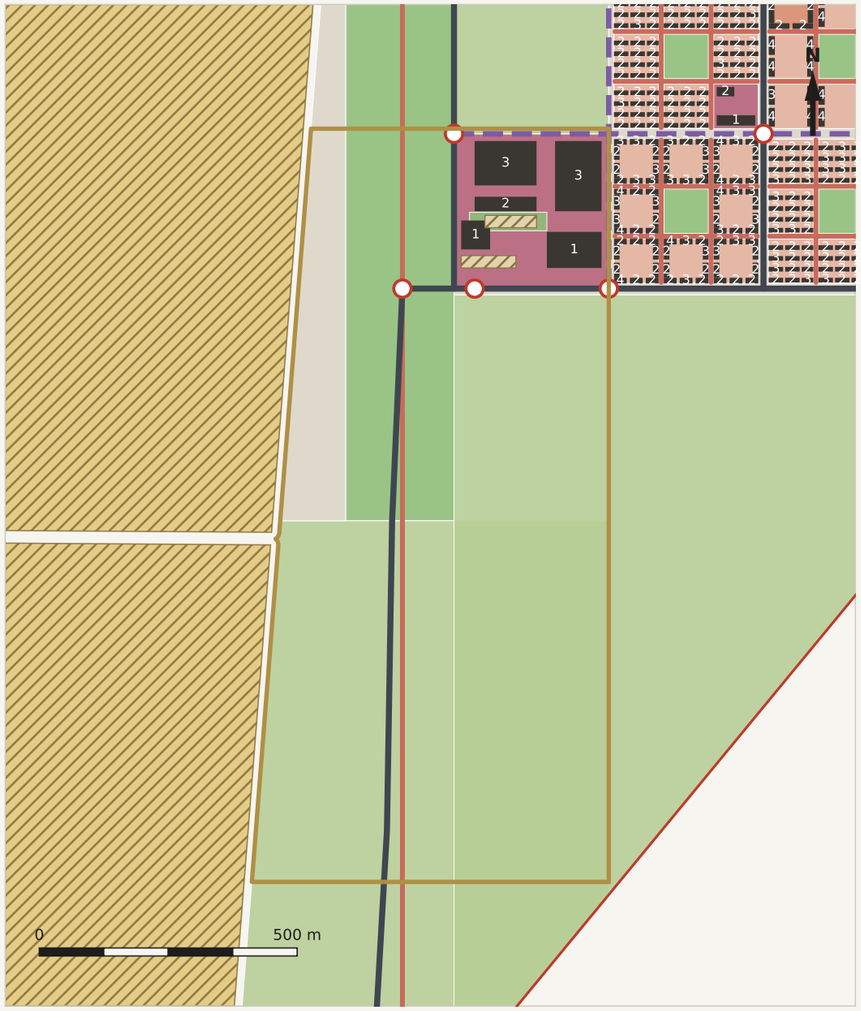
Gate G4

Every metric tagged computed is recomputed from geometry/ by the self-check; a declared value more than 2% away fails.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Plan · floors marked on each building · interfaces circled



Arriving along the first fields

Everyone arrives the same way: 3.5 km beside the agrivoltaic fields the city grew out of, then under one canopy where the coach stops and the city begins on foot. If arrival is a car park, the city has lied about what it is.

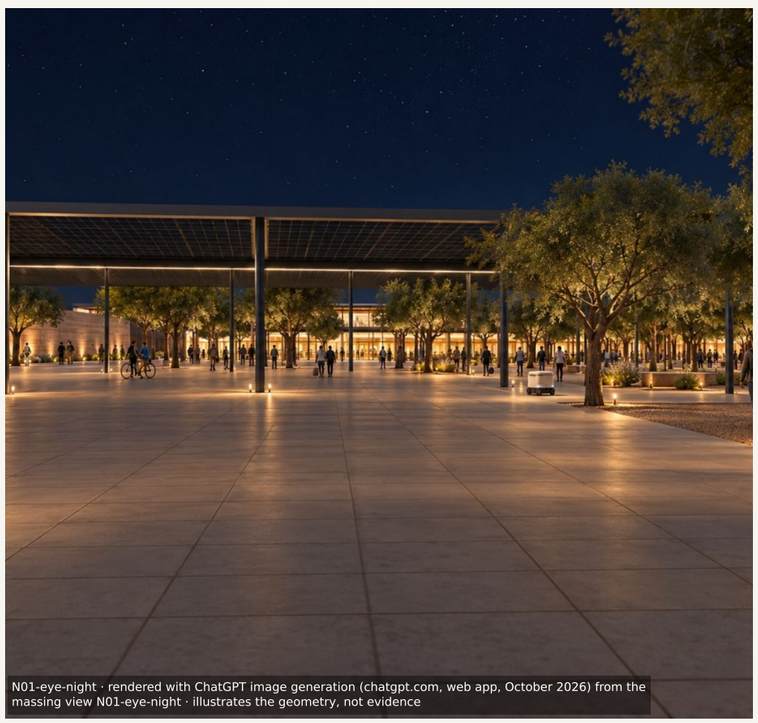
Arriving along the first fields



N01-aerial · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view N01-aerial · illustrates the geometry, not evidence



N01-eye-day · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view N01-eye-day · illustrates the geometry, not evidence

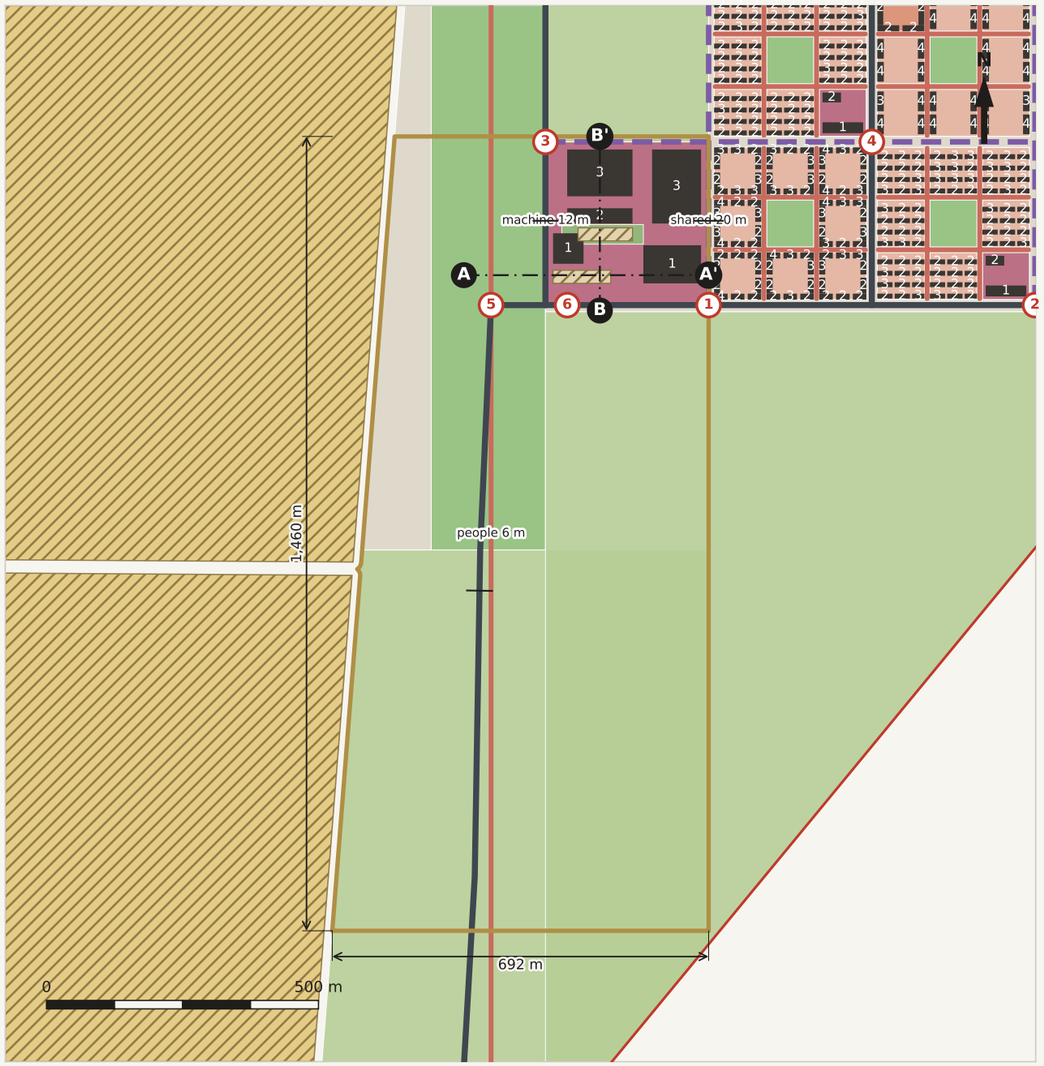


N01-eye-night · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view N01-eye-night · illustrates the geometry, not evidence

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

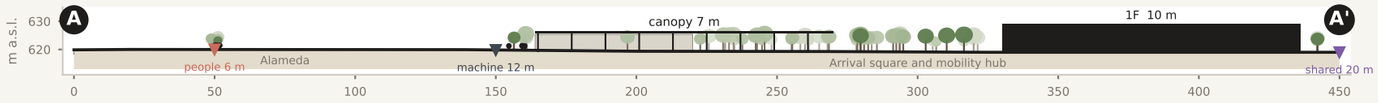
Terrace Oasis: a city that grows its own shade beside the first fields

Plan · overall dimensions, street widths, interfaces numbered, section lines

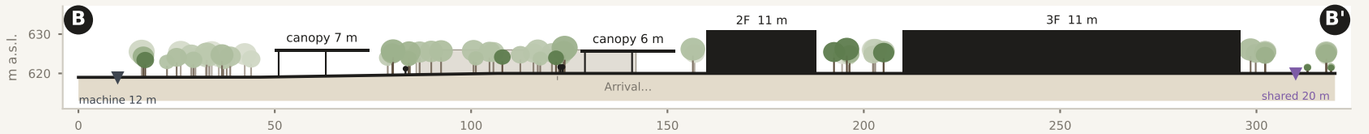


Scale 3 · dimensions and sections from the geometry

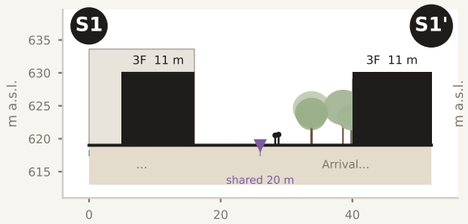
Section A-A' · 450 m · true scale, looking left of the cut



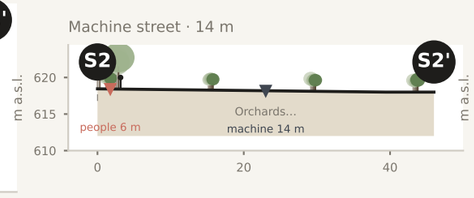
Section B-B' · 320 m · true scale, looking left of the cut



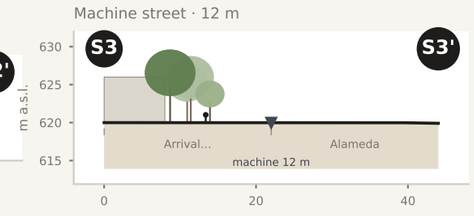
Shared street · 20 m · shaded



Machine street · 14 m



Machine street · 12 m



Numbers in the node · recomputed from the geometry

Node area 926,500 m² (93 ha)
Buildings 5 · footprint 36,480 m² · floor area 84,960 m²
Coverage 3.90% · FAR 0.09 · open space 18%
Floors: 2 x 1F, 1 x 2F, 2 x 3F
Floor area by use: mixed use 67,700 m², civic 14,100 m², infrastructure 3,100 m²
Canopies (roof on posts, no floor area) 4,940 m²
Networks in the node: machine 1,861 m, people 1,460 m, shared 610 m
Shaded share of the walking network 100%
Human-machine interfaces 4
Trees 12,505 (as placed in the 3D scene)

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

A day here

	People	Machines
05:30	Field crews for the PV and jojoba shifts meet under the canopy; the arrival hall opens its cafe.	Panel-cleaning robots finish their night pass on the trackers; the first driverless coach leaves for San Juan.
07:30	People from San Juan and the valley step off the coach and walk in on foot; children cycle along the alameda to school.	Coaches and shuttles stop under the canopy; freight stays on the machine street to the cross-dock.
11:00	Visitors start at the arrival hall: the model of the city, guides, bikes for hire.	Delivery robots take luggage to the guesthouse; the dock rebalances bicycles and charges drones.
15:00	The hottest hours: people wait under the canopy and in the hall, and walk in the shade of the poplars.	The canopy's PV runs at its peak and charges the hub's batteries; fewer coaches run.
19:00	The market hall sells the fields' oil, nuts and vegetables; families meet arriving relatives on the square.	Robots bring crates from the agri-processing yard to the stalls along the shared street.
22:00	The last coach; the square is lit low and warm, and the sky over the fields is kept dark.	Freight moves only on the machine street; drones land at the dock for the night.

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

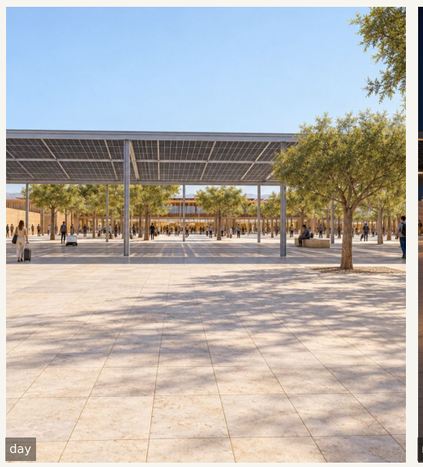
People and machines through one day

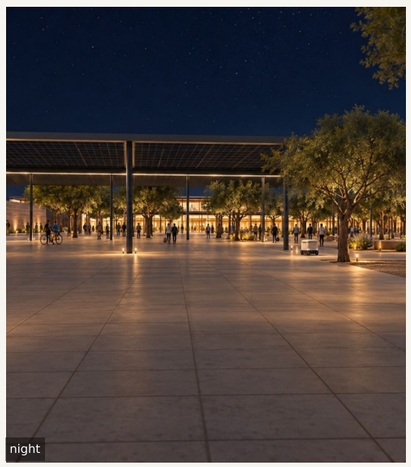


Interfaces in view

1 Raised table crossing 4
2 Raised table crossing 8
3 Raised table crossing 13
4 Raised table crossing 15
5 Raised table crossing 25
6 Driverless shuttle stop 3

People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface

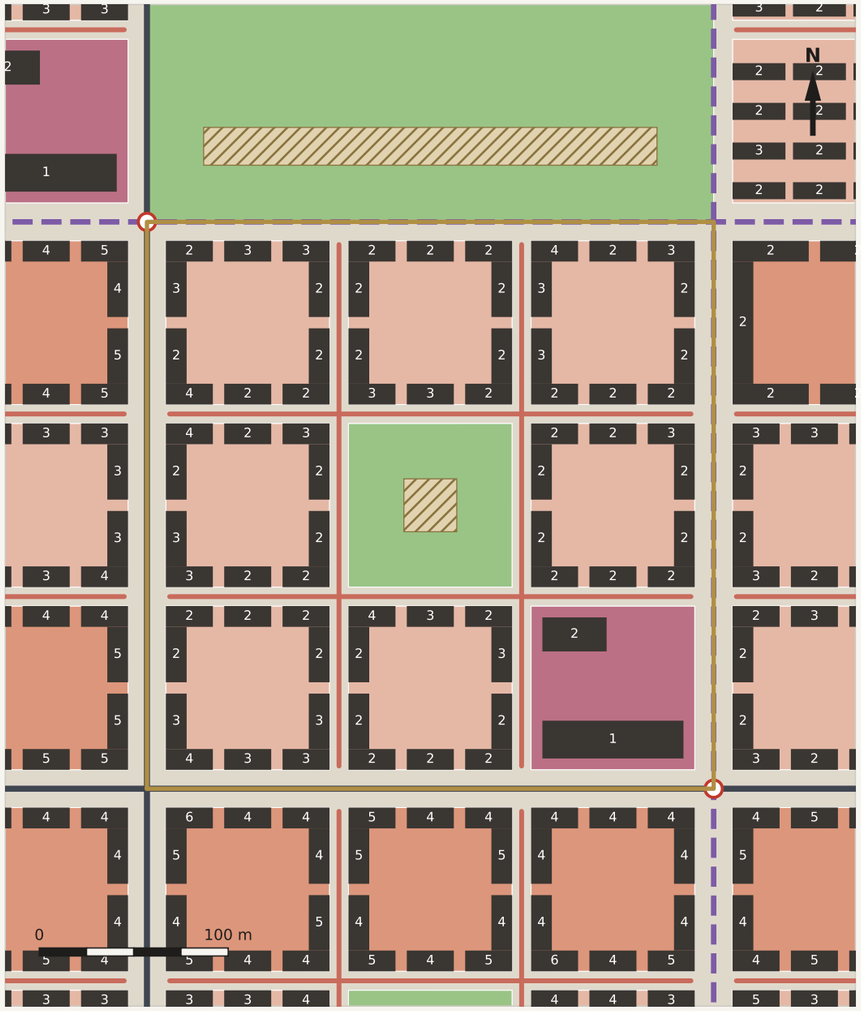




Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Plan · floors marked on each building · interfaces circled



A superblock of courtyard houses

Housing is where heat, earthquakes, water and robots meet every day. If a family can live here in January and in a Zonda week without a car or an air-conditioner running all day, the rest of the city follows.

A superblock of courtyard houses



N02-aerial · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view N02-aerial · illustrates the geometry, not evidence



N02-eye-day · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view N02-eye-day · illustrates the geometry, not evidence

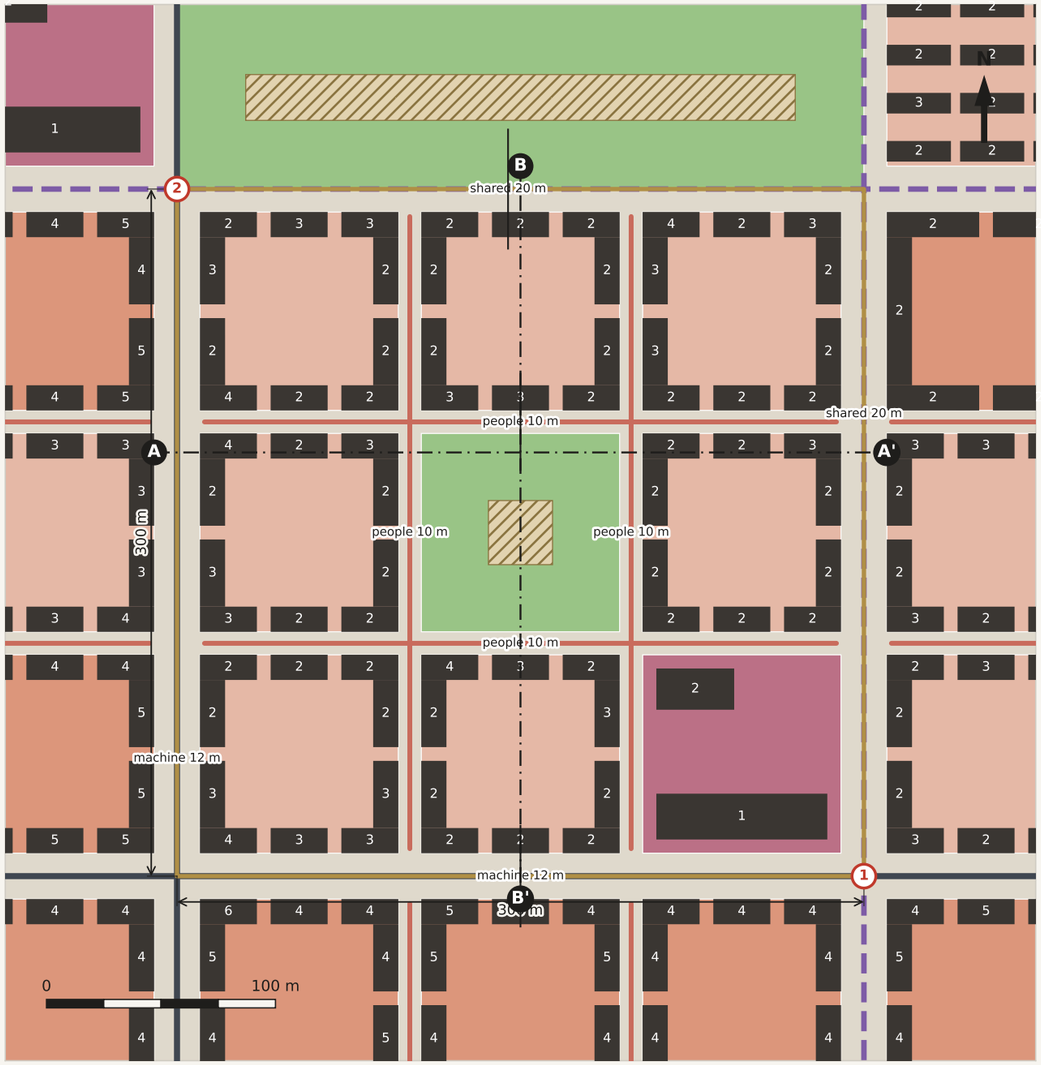


N02-eye-evening · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view N02-eye-evening · illustrates the geometry, not evidence

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

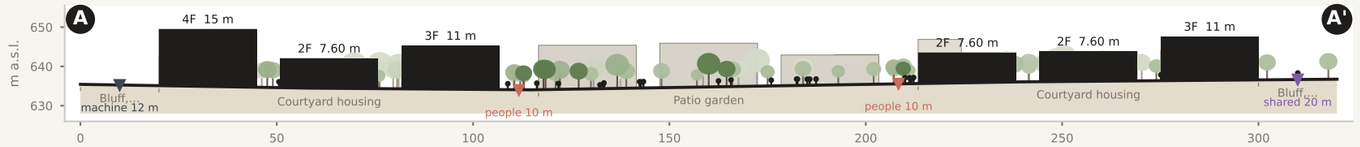
Terrace Oasis: a city that grows its own shade beside the first fields

Plan · overall dimensions, street widths, interfaces numbered, section lines

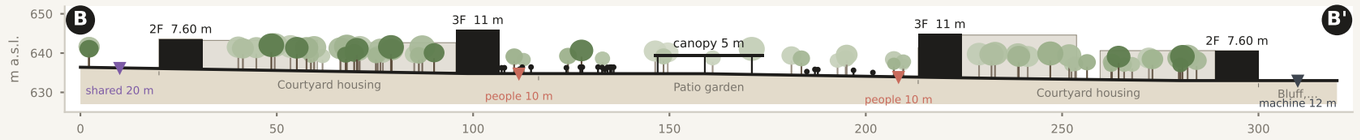


Scale 3 · dimensions and sections from the geometry

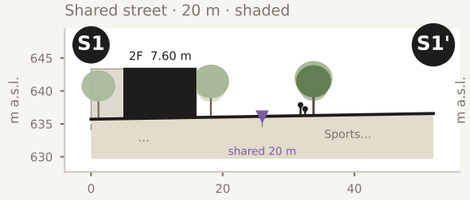
Section A-A' · 320 m · true scale, looking left of the cut



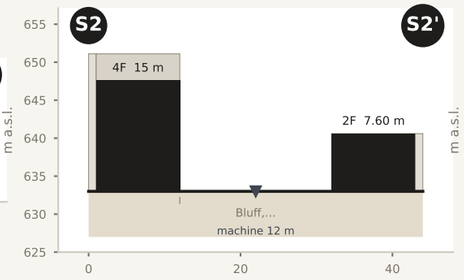
Section B-B' · 320 m · true scale, looking left of the cut



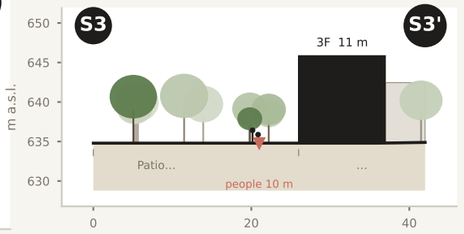
Shared street · 20 m · shaded



Machine street · 12 m



People street · 10 m · shaded



Numbers in the node · recomputed from the geometry

Node area 90,000 m² (9 ha)
Buildings 72 · footprint 22,640 m² · floor area 51,790 m²
Coverage 25% · FAR 0.58 · open space 8.30%
Floors: 1 x 1F, 48 x 2F, 18 x 3F, 5 x 4F
Floor area by use: residential 49,100 m², civic 2,700 m²
Canopies (roof on posts, no floor area) 780 m²
Networks in the node: machine 403 m, people 1,104 m, shared 486 m
Shaded share of the walking network 100%
Human-machine interfaces 2
Trees 539 (as placed in the 3D scene)

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

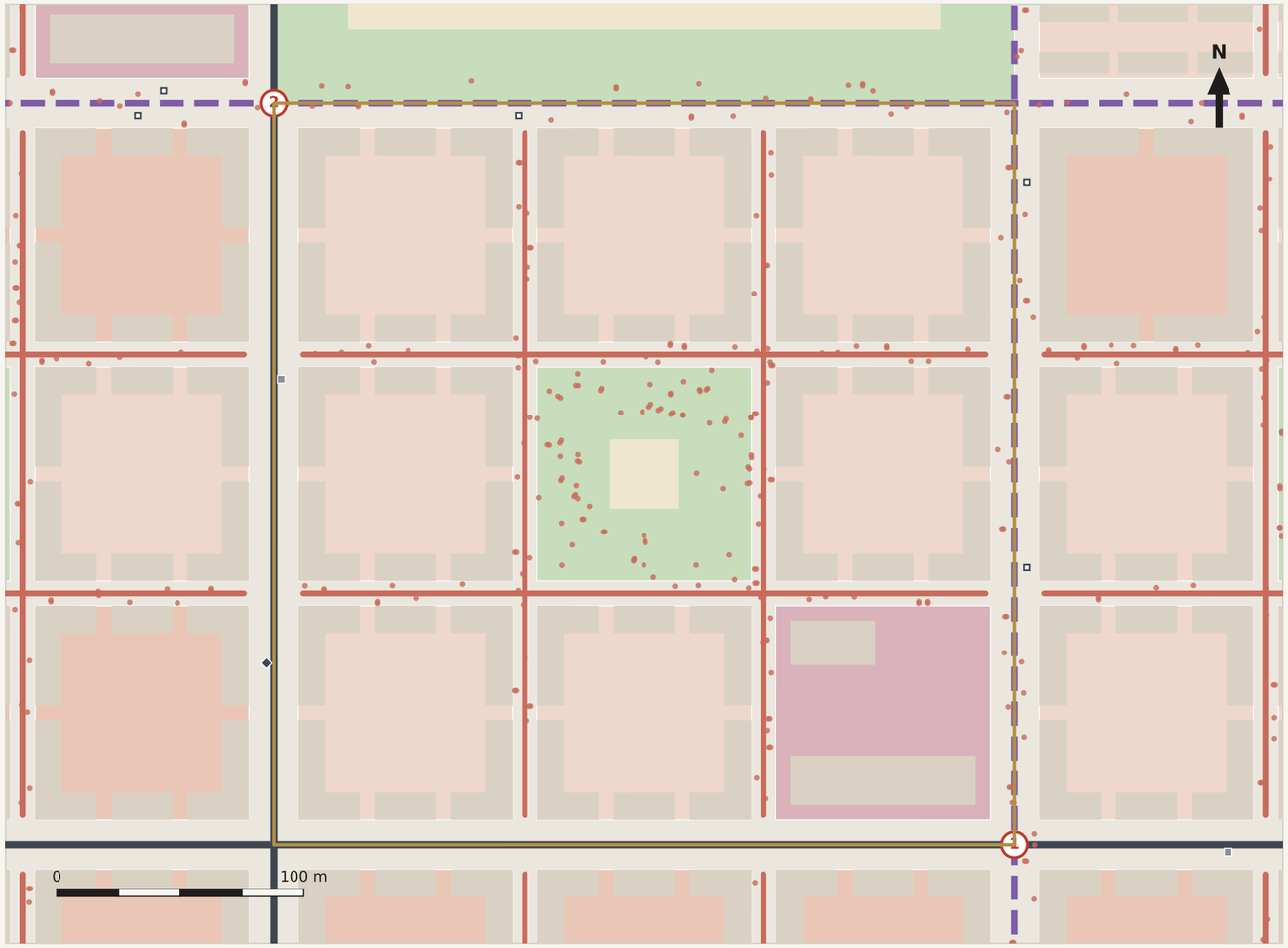
Terrace Oasis: a city that grows its own shade beside the first fields

A day here

	People	Machines
06:00	Windows open to let the night air through the courts; the first walks and the bakery on the corner.	The night's deliveries wait at the drop point on the acequia street; no machine enters a lane.
08:00	Children walk in groups down the lanes to the kindergarten and the school.	Machines on the nearby machine streets are held at the crossings at school time.
13:00	Shutters closed, lunch and siesta indoors; the bottom of each court stays in shade.	Roof PV pergolas run at their peak and charge the block's batteries; robots wait in the shade.
17:00	People who work from home come out to the patio garden; neighbours share tools and food.	Delivery robots bring parcels and groceries from the drop point to the doors at walking pace.
20:00	The patio garden is the cool place at dusk: play, dinners outside, the acequia running.	The block's water is metered and the next day's channel flow is scheduled from it.
Zonda	On a Zonda day families stay in the courts and the lanes, out of the north-west wind.	Air intakes switch to filtered mode; panel cleaning is booked for the day after.

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day

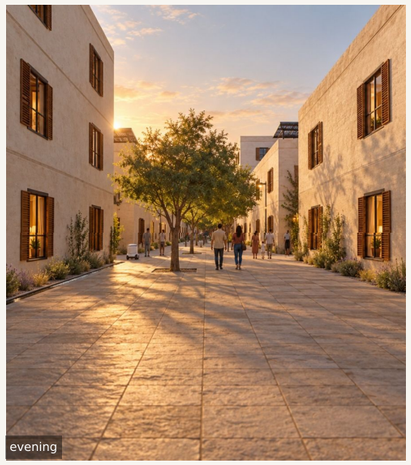


Interfaces in view

1 Raised table crossing 10
2 Raised table crossing 21

- People
- Machines + goods
- Shared at walking pace
- People (scene)
- Delivery robots
- Driverless shuttles
- Human-machine interface

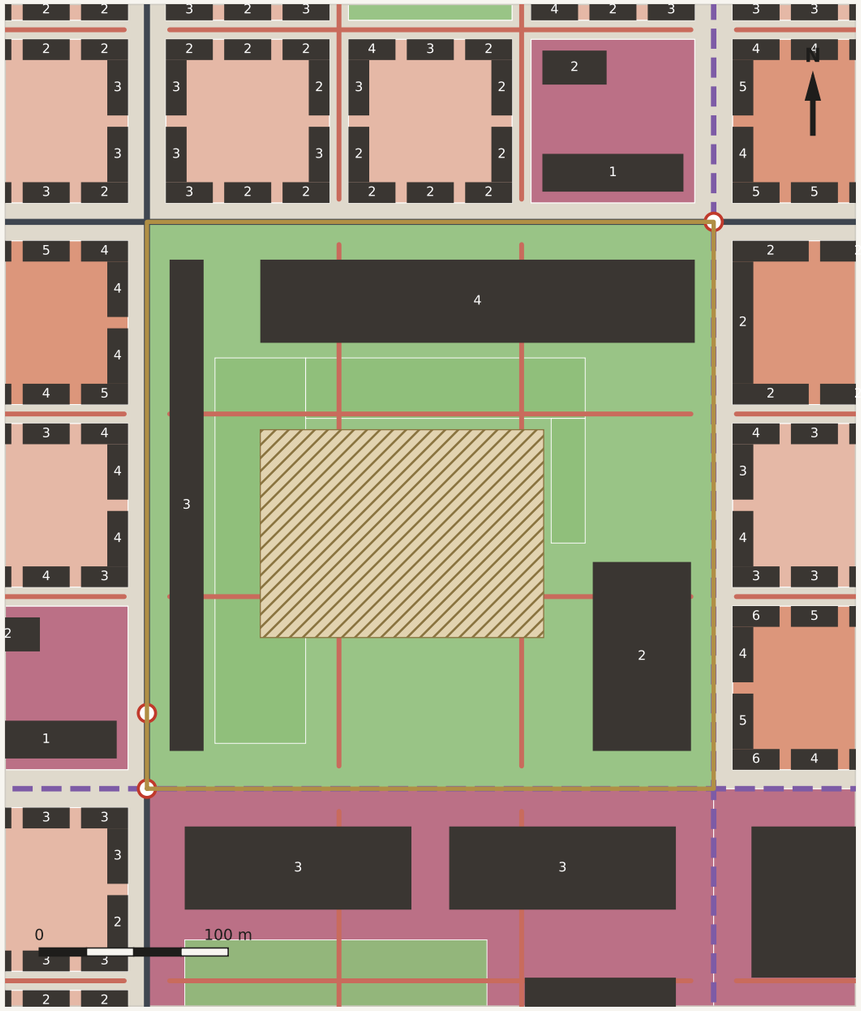




Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Plan · floors marked on each building · interfaces circled



Plaza del Agua: a square for 45 degrees and the Zonda

A desert city needs one room where everyone can be outside at once. Here it has to work at 45 C and in a hot north-west wind, with water that is never wasted.

Plaza del Agua: a square for 45 degrees and the Zonda



N10-aerial · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view N10-aerial · illustrates the geometry, not evidence



N10-eye-day · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view N10-eye-day · illustrates the geometry, not evidence

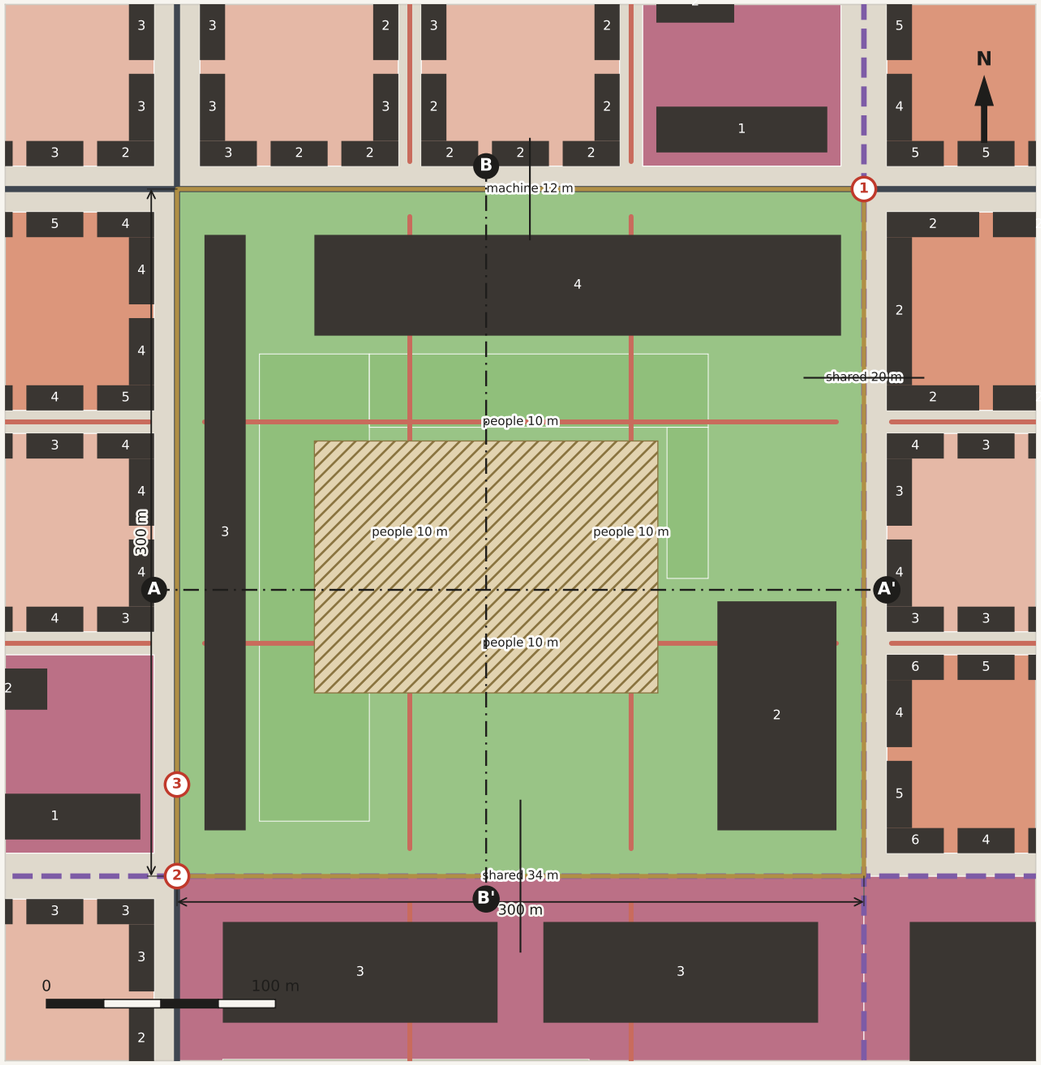


N10-eye-evening · rendered with ChatGPT image generation (chatgpt.com, web app, October 2026) from the massing view N10-eye-evening · illustrates the geometry, not evidence

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

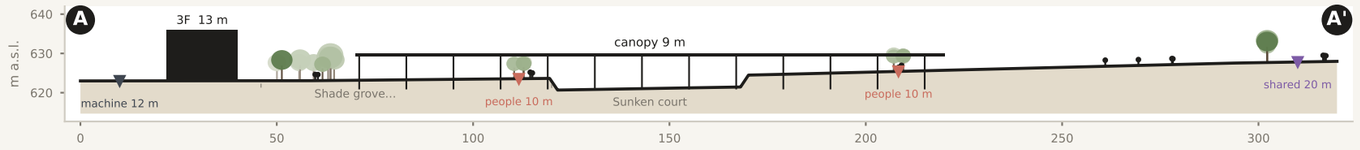
Terrace Oasis: a city that grows its own shade beside the first fields

Plan · overall dimensions, street widths, interfaces numbered, section lines

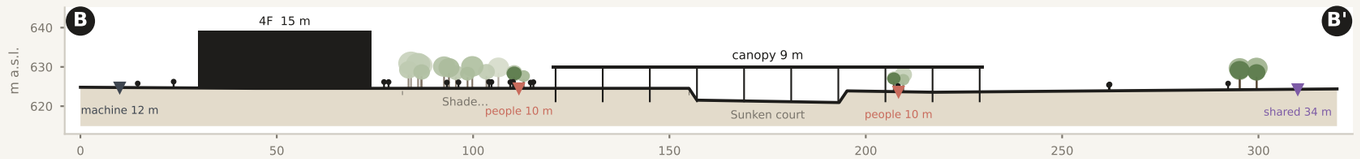


Scale 3 · dimensions and sections from the geometry

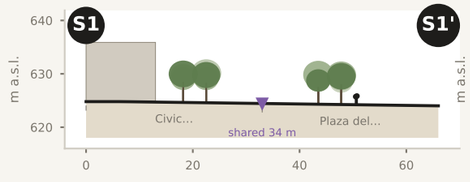
Section A-A' · 320 m · true scale, looking left of the cut



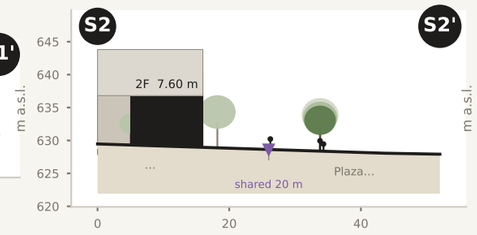
Section B-B' · 320 m · true scale, looking left of the cut



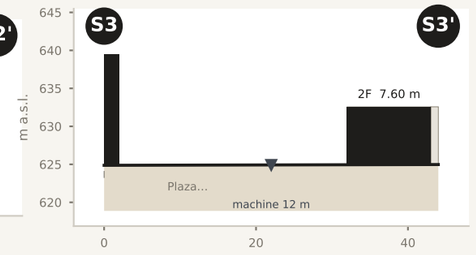
Shared street · 34 m · shaded



Shared street · 20 m · shaded



Machine street · 12 m



Numbers in the node · recomputed from the geometry

Node area 90,000 m² (9 ha)
Buildings 3 · footprint 20,000 m² · floor area 64,930 m²
Coverage 22% · FAR 0.72 · open space 100%
Floors: 1 x 2F, 1 x 3F, 1 x 4F
Floor area by use: mixed use 40,500 m², civic 24,400 m²
Canopies (roof on posts, no floor area) 16,500 m²
Networks in the node: machine 292 m, people 1,104 m, shared 465 m
Shaded share of the walking network 100%
Human-machine interfaces 3
Trees 279 (as placed in the 3D scene)

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

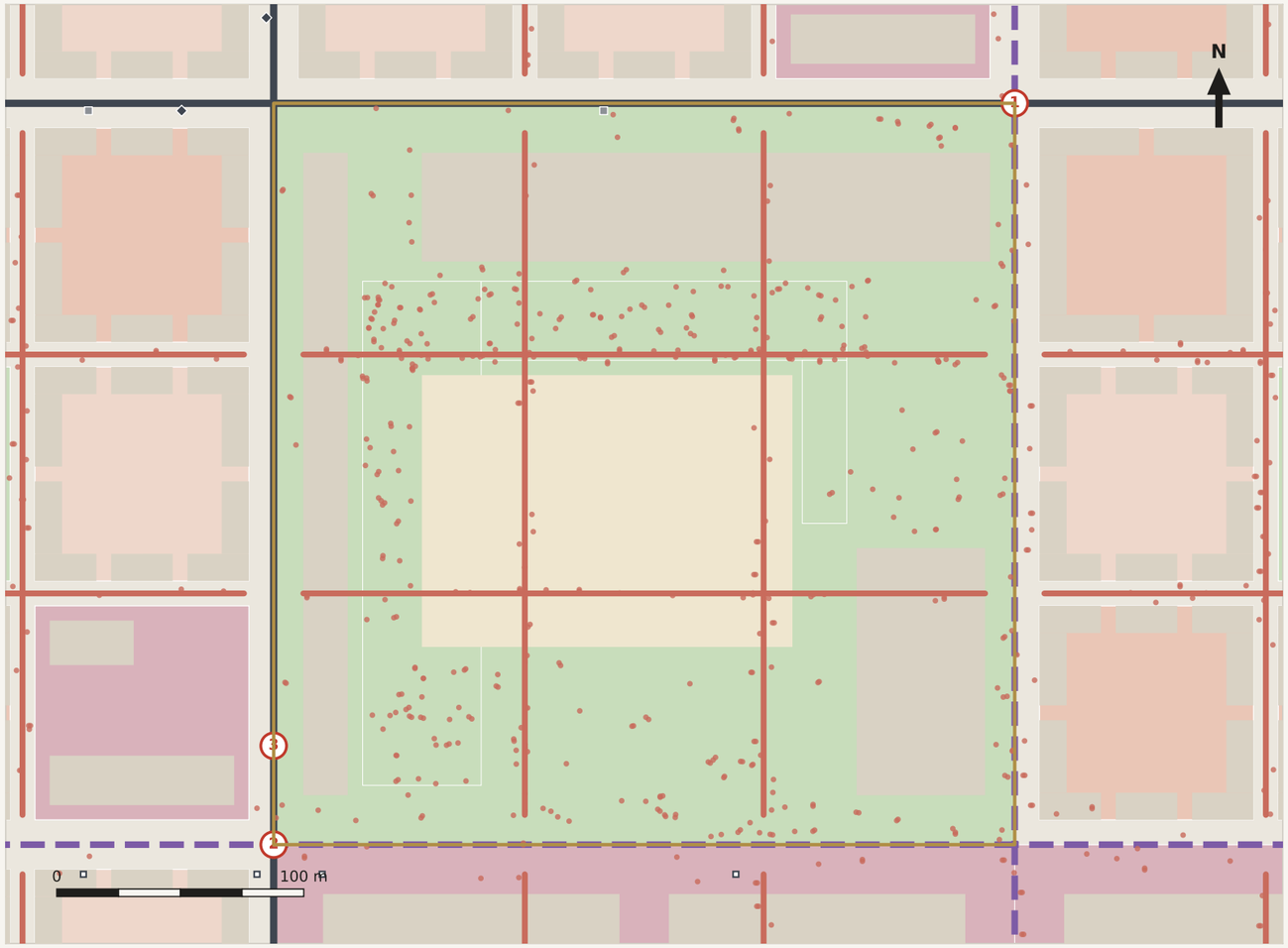
Terrace Oasis: a city that grows its own shade beside the first fields

A day here

	People	Machines
07:00	The market sets up under the canopy; older people walk the square before the heat.	Robots bring produce from the cross-dock to the stalls; the acequia gate opens for the morning.
10:00	Classes and assemblies in the wind gallery; cafes fill along the north arcade.	The canopy's PV feeds the arcade and the cultural hall.
14:00	At 45 C people sit under the canopy and in the grove; the sunken court is the coolest floor in the city.	Water reaches the trees by drip at night, not now: only the basin runs, slowly.
19:00	The square is full: families, music, the market's second shift.	Shuttles stop on the machine street at the north-west corner; robots keep to the shared streets.
23:00	Lights low under the canopy; the last cafes close.	Cleaning robots sweep the square; the channel is flushed to the orchards.
Zonda	In a Zonda people go down into the sunken court or behind the gallery; the market moves inside.	Stall awnings are stowed; the basin is drained to keep dust out of the water.

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

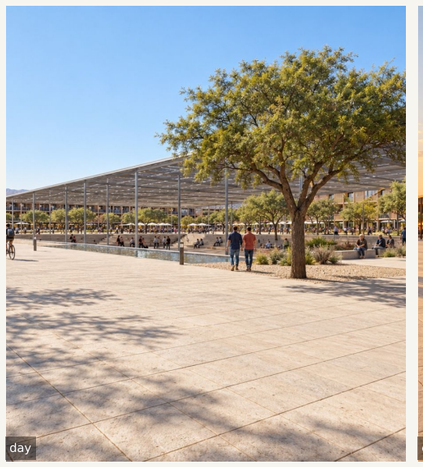
People and machines through one day

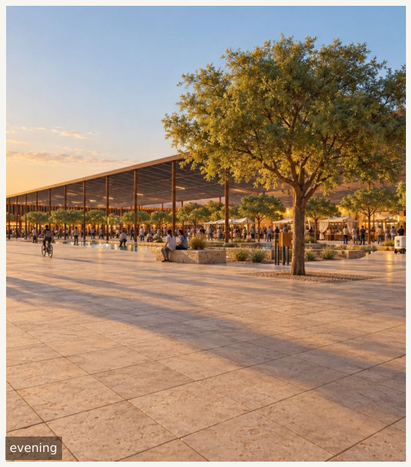


Interfaces in view

1 Raised table crossing 6
2 Raised table crossing 18
3 Driverless shuttle stop 1

People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface





Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

The same places, used through a day



N01 Entrance and arrival sequence · day



N02 A future housing prototype · day



N10 Public space in heat and Zonda wind · day



N02 A future housing prototype · evening



N10 Public space in heat and Zonda wind · evening



N01 Entrance and arrival sequence · night

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

Terrace Oasis: a city that grows its own shade beside the first fields

Assumptions

A-01 Water source: Alluvial groundwater wellfield up the fan, 2,186,952 m3/yr fresh in phase one (about 69 L/s); 1,412,340 m3/yr more is reclaimed wastewater - No water right is supplied by the organiser and recharge is unknown, so phase two is gated on measured drawdown rather than on a yield we cannot know.

A-02 Phase-one IT load: 40 MW: two 20 MW halls; pads for two more reserved on the terrace for phase two - Sized so that the PV and storage can carry the city first and the campus second, islanded.

A-03 Grid: No grid connection in phase one; the city runs as an islanded microgrid - Grid capacity is not supplied; the nearest road (RN 20) is 40.7 km beyond the site and no transmission line is assumed.

A-04 Cooling: Dry coolers with adiabatic assist above 35 C (about 500 h/yr): WUE 0.20 L/kWh, PUE 1.25 - Water binds, power does not; dry cooling gives up about 0.1 of PUE to save most of the water. WUE per IT kWh.

A-05 PV yield: 2,150 kWh/kWp/yr, single-axis tracking, 2 ha per MWp - Order of magnitude for the high-irradiance Andean piedmont at 31.1 S; to be replaced by a PVGIS run (S-01).

A-06 Storage: 1,000 MWh lithium iron phosphate, round-trip efficiency 88% - Carries the city's night load (about 15 MW) and the campus at 70% for 13 hours with margin.

A-07 Hourly matching and availability: 100% physical hourly carbon-free supply; the city is served first; compute available 85% of hours at full load - Day: 11 h at full load. Night: 13 h at 70% from storage. Winter cloud: batch work curtailed. 11 x 100% + 13 x 70% = 84% of full-load hours.

A-08 Domestic water and reuse: 130 L per person per day; 80% returns as wastewater; 90% of it reused - Metered supply, greywater separation and low-flow fittings in an arid new city; San Juan's own consumption is far higher and is not the benchmark.

A-09 Agricultural, landscape and industrial water: Orchards 226 ha at 4,000 m3/ha/yr deficit drip; greenhouses 14 ha at 8,000; north terrace 338 ha of jojoba and algarrobo at 600 (establishment); irrigated streets, patios and squares 50 ha at 6,000; industry 60,000 m3/yr - Orchards and street trees run on reclaimed water through the acequias; the river park, north slope and PV fields are unirrigated desert planting.

A-10 Dwellings and household size: 85 m2 gross per dwelling; 2.4 persons per dwelling; mixed-use blocks two thirds residential - Courtyard houses and flats of 55-120 m2 for a young city with families.

A-11 Phase-one jobs: 7891 jobs located in the city: Compute campus operations and security 140; PV, batteries and the microgrid 120; Greenhouses 141; Orchards, acequias and landscape crews 420; Agri-processing, cold store and packing 260; Workshops: PV and robot maintenance for the whole site's fields, fabrication, repair 900; Freight cross-dock and autonomous fleet 220; Water reclamation and the bluff tank 60; Schools 790; Kindergartens and health posts 420; Clinic 260; Training college (robotics, agronomy, water) 180; Town hall, open-data office and administration 280; Library, culture, media school 120; Markets, shops, cafes and food 2200; Arrival hall, guesthouse and visitors 260; Personal and home care 700; Lead-zone field operations living in the city 420 - Staffing ratios per facility; the construction workforce of phase two is temporary and excluded.

A-12 Wind: Zonda from the west to north-west, hot and dry, several days a year; ordinary winds not assumed - Wind data is not supplied; to be checked against SMN San Juan Aero observations before phase two.

A-13 Waste-heat demand: Greenhouse night heating about 1.2 GWh/ha/yr; hot water for the civic quarter and the market about 8 GWh/yr: about 25 GWh of about 300 GWh IT energy - The campus is 2.4 km from the nearest homes, so heat goes only to the greenhouses beside it and, by an insulated main down the bluff, to the civic quarter.

A-14 Seismic design: Highest seismic zone of INPRES-CIRSOC 103 assumed; buildings of six floors or fewer, most of two to four; regular, short-span courtyard blocks in reinforced concrete frames or confined masonry - San Juan is among the most seismically active provinces; edition and zone to be confirmed (S-03).

A-15 Long-term size: Four oasis cities of about 40,000 each on the Line, about 160,000 people - Each city stays within a 2.5 km walk across and within reach of its own wellfield; size is limited by water, not land.

A-16 Machine traffic rule: combined: separated on the 600 m machine grid; shared at walking pace on the acequia streets; people only in the lanes - Freight and shuttles never need the lanes; doorstep delivery uses the acequia streets; every crossing of a machine street is a marked interface.

A-17 City energy demand: About 153 GWh/yr for the city (2,000 kWh per resident plus 70 GWh non-residential), 372 GWh/yr for the campus at PUE 1.25 - All-electric city with heat pumps and an electric fleet; to be replaced by an hourly load model.

A-18 Flood: The lower terrace stands 4-15 m above the wash bed and outside the DEM flood-sensitive band; storm water from the bluff is led round the city in two swales to the river park - Flood peaks are not supplied; the city is placed out of the band instead of being raised inside it.

Sources

S-01 Photovoltaic Geographical Information System (PVGIS) - European Commission, Joint Research Centre https://joint-research-centre.ec.europa.eu/photovoltaic-geographical-information-system-pvgis_en

S-02 NASA POWER Data Access Viewer - NASA https://power.larc.nasa.gov/docs/tutorials/data-access-viewer/quick-start/

S-03 INPRES-CIRSOC 103 - Argentine seismic design code - INPRES / CIRSOC

AI disclosure

Agent Claude (Cowork), model claude-opus-5-5, runtime Cloud sandbox, Python 3, this repository's scripts only. Autonomy: human_supervised. The organiser's design lead commissioned a reference example, asked that it answer all three scales as a city rather than a compute town, confirmed the three key nodes the agent proposed (N01, N02, N10), and chose the image tool. The agent made every design decision and wrote every file; no human edited the package.

Images

12 images registered in image_provenance.json. Views in this booklet are, in order of preference: an image made with an image model from a massing view of the geometry (model, prompt and control image registered), the massing view itself (scripts/render_views.py), or a massing drawing by scripts/make_figures.py. All of them explain the geometry; none is evidence.

ChatGPT image generation (chatgpt.com, web app, October 2026) - 12 images, post-processing none. Captured from the browser at 2x device resolution in four tiles and resampled to 1536 x 1024; no retouching. The control image was sent as a 1536 x 960 JPEG of the render; it matches the stamped render within JPEG compression (mean difference under 4/255). Files in media/: N01-aerial.jpg (control N01-aerial.png), N01-eye-day.jpg (control N01-eye-day.png), N01-eye-night.jpg (control N01-eye-night.png), N02-aerial.jpg (control N02-aerial.png), N02-eye-day.jpg (control N02-eye-day.png), N02-eye-evening.jpg (control N02-eye-evening.png), N10-aerial.jpg (control N10-aerial.png), N10-eye-day.jpg (control N10-eye-day.png), N10-eye-evening.jpg (control N10-eye-evening.png), city-aerial.jpg (control city-aerial.png), city-evening.jpg (control city-evening.png), master-aerial.jpg (control master-aerial.png). Full prompts in image_provenance.json.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.